# Correlation Network Studio

Interactive Graph Analytics, Statistical Modeling & High-Dimensional Feature Exploration

**Full-Stack System: FastAPI • SQLite • D3.js Force Physics Simulation • SciPy & NumPy Analytics**

Academic Project Defense • Comprehensive Evaluation: Real Datasets, Empirical Regressions & Centrality Topology

# Preface & Objective

**The Challenge:** High-dimensional tabular datasets obscure cross-variable correlations. Traditional N × N correlation tables scale quadratically ($O(N^2)$), overwhelming human cognitive bandwidth. Raw numerical tables hide latent clusters, hub variables, and inverse trade-offs.

**Our Objective:** To engineer and deploy a production full-stack visual analytics platform transforming raw numerical datasets into interactive, force-directed network graphs with linked bivariate OLS regression scatter plots and univariate distribution metrics.

**The Scope:** This presentation outlines our decoupled three-tier architecture (FastAPI, SQLite, D3.js), the mathematical correlation engines (Pearson & Spearman with two-tailed p-values), empirical dataset evaluations, and multi-level analytical user workflows.

SYSTEM WORKFLOW

# Methodology: The Core Pipeline

Three integrated analytical stages transforming raw tabular data into interactive visual insights.

## 1. Data Ingestion & Storage

**STEP 1: PERSISTENCE**

- Aggregates and persists multi-feature datasets in SQLite relational database.
- Dynamic schema validation requires ≥ 2 numeric columns and ≥ 3 observations.
- Automated drag-and-drop CSV uploader with Pandas numeric type sanitization.
- Pre-seeded with real benchmark datasets: Academic Performance & Macroeconomics.
- UUID generation and transactional integrity guarantee reliable data retrieval.

## 2. Statistical Engine Core

**STEP 2: ANALYTICS**

- Dual correlation engine: computes full N × N Pearson and Spearman rank matrices.
- Rigorous hypothesis testing: calculates exact two-tailed Student's t-test p-values.
- Dynamic threshold filtering: strips spurious, statistically non-significant edges.
- Bivariate OLS linear regression: fits trendline, calculates slope, intercept, and $R^2$.
- Univariate statistics: NumPy frequency histogram binning and quartile metrics.

## 3. Interactive Visual Analytics

**STEP 3: VISUALIZATION**

- D3.js v7 force physics engine with continuous electrostatic repulsion and spring links.
- Visual encodings: node radius ∝ weighted degree centrality; edge color/dash ∝ polarity.
- Symmetric correlation heatmap with diverging rust-to-teal gradient ramp.
- Linked drill-down modals: click edge → 2D scatter plot; click node → histogram.
- Publication export: one-click vector SVG graphic download.

**Key Takeaway:** The end-to-end pipeline operates with sub-50ms roundtrip latency, providing real-time feedback as analysts adjust significance and correlation thresholds.

SYSTEM ARCHITECTURE

# Three-Tier System Architecture & Data Flow

Decoupled full-stack architecture bridging Python scientific computation with web-native D3.js physics.

## Frontend Presentation Tier

**PRESENTATION**

- Pure ES6 modular architecture (api.js, graph.js, heatmap.js, charts.js, app.js). Zero bundler overhead.

## D3.js Force Physics Engine

**SIMULATION**

- Velocity Verlet integration balancing electrostatic repulsion (charge = -320), link springs, and collisions.

## FastAPI Application Tier

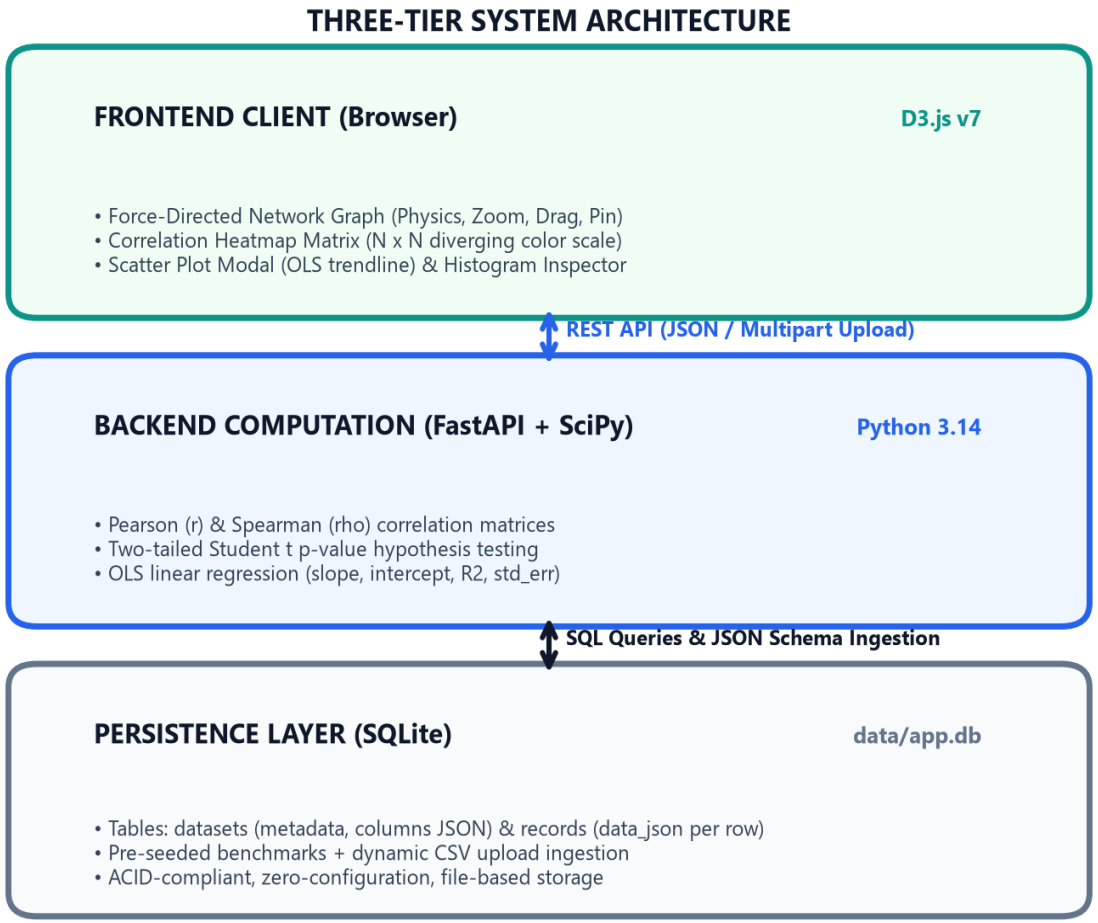
**BACKEND API**

- High-throughput ASGI server exposing RESTful endpoints with Pydantic validation and auto Swagger UI.

## SQLite Persistence & Math Core

**DATA & MATH CORE**

- ACID-compliant storage paired with SciPy/NumPy for $O(N^2)$ correlation and OLS regression modeling.



STATISTICAL ENGINE

# Mathematical & Algorithmic Foundations

Rigorous statistical formulations implemented in SciPy and NumPy for dependable relationship inference.

## Pearson Linear Correlation (r)

**PARAMETRIC METRIC**

- Formula: $r = \Sigma((x_i - \bar{x})(y_i - \bar{y})) / [\sqrt{\Sigma(x_i - \bar{x})^2}\ \sqrt{\Sigma(y_i - \bar{y})^2}] = Cov(X,Y) / (\sigma_x \cdot \sigma_y)$
- Measures the degree and direction of linear co-variation between continuous variables; bounds: [-1, +1].
- In Studio: Drives primary spring link attraction and edge stroke width.

## Spearman Rank Correlation (ρ)

**NON-PARAMETRIC METRIC**

- Formula: $\rho = 1 - [6\ \Sigma\ d_i^2] / [n(n^2 - 1)]$, where $d_i = rank(x_i) - rank(y_i)$
- Non-parametric metric evaluating monotonic relationships regardless of whether they are linear.
- In Studio: Robust against extreme outliers and skewed distributions; selectable via UI toggle.

## Two-Tailed Hypothesis Testing (p-value)

**SIGNIFICANCE VALIDATION**

- Test Statistic: $t = r \cdot \sqrt{[(n - 2) / (1 - r^2)]}$ with degrees of freedom $df = n - 2$
- Exact p-value computed via Student's t cumulative distribution function: $p = 2 \cdot (1 - T_cdf(|t|, df))$.
- In Studio: Real-time significance filter removes spurious edges with $p \geq 0.05$, preventing false discoveries.

## Ordinary Least Squares (OLS) Regression

**PREDICTIVE MODELING**

- Model: $\hat{y} = \beta_1 x + \beta_o$, where $\beta_1 = Cov(X,Y)/Var(X)$ and $\beta_o = \bar{y} - \beta_1 \bar{x}$
- Coefficient of Determination: $R^2 = 1 - (SS_res / SS_tot) = r^2$ (proportion of variance explained).
- In Studio: Modal scatter plot dynamically fits regression line and displays equation with $R^2$ and p-value.

EMPIRICAL EVALUATION

# Benchmark Dataset 1: Student Academic Performance (N=100)

8 academic subjects, N=100 observations, 28 total pairwise correlations with latent multi-factor clustering.

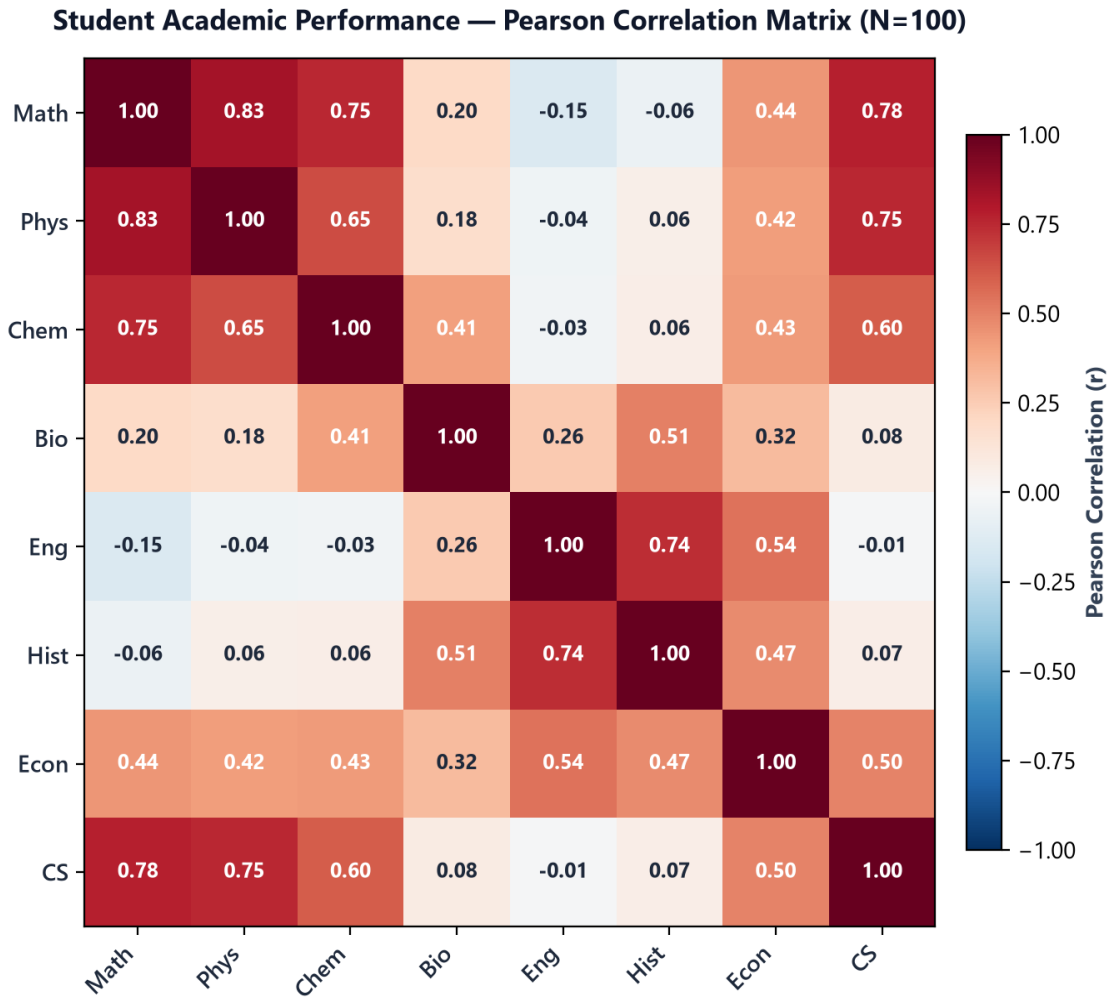
| Variable | Mean | Std Dev | Range [Min-Max] | Median | Weighted Degree | Latent Cluster |
|---|---|---|---|---|---|---|
| **Mathematics** | 70.12 | 10.42 | 45.0 - 92.4 | 69.80 | **3.214 (Rank 1 - Hub)** | STEM Core Cluster |
| **Economics** | 70.31 | 11.51 | 40.2 - 96.1 | 70.15 | **3.119 (Rank 2 - Hub)** | Quant / Verbal Bridge |
| **Chemistry** | 68.84 | 9.78 | 44.1 - 90.3 | 68.50 | **2.937 (Rank 3)** | STEM / Life Science |
| **Computer Science** | 72.45 | 11.23 | 45.3 - 98.0 | 72.10 | **2.827 (Rank 4)** | STEM Core Cluster |
| **Physics** | 69.21 | 10.89 | 42.0 - 94.2 | 68.90 | **2.766 (Rank 5)** | STEM Core Cluster |
| **History** | 69.82 | 10.12 | 46.1 - 95.0 | 69.40 | **2.112 (Rank 6)** | Humanities Cluster |
| **English** | 71.53 | 9.64 | 48.0 - 93.2 | 71.20 | **2.051 (Rank 7)** | Humanities Cluster |
| **Biology** | 67.91 | 10.81 | 38.5 - 91.0 | 67.80 | **1.842 (Rank 8)** | Life Sciences Core |

**Key Takeaway:** Generative Architecture: Combines 4 latent factors: Quantitative (STEM), Verbal (Humanities), Memory (Life Science), and Applied Reasoning (Economics). Mathematics and Economics emerge as the primary network hubs with the highest weighted degree centrality.

CORRELATION MATRIX

# Dataset 1: Full 8×8 Correlation Heatmap Analysis

Symmetric correlation matrix revealing distinct disciplinary clusters, cross-domain bridges, and significance patterns.




## Empirical Matrix Insights

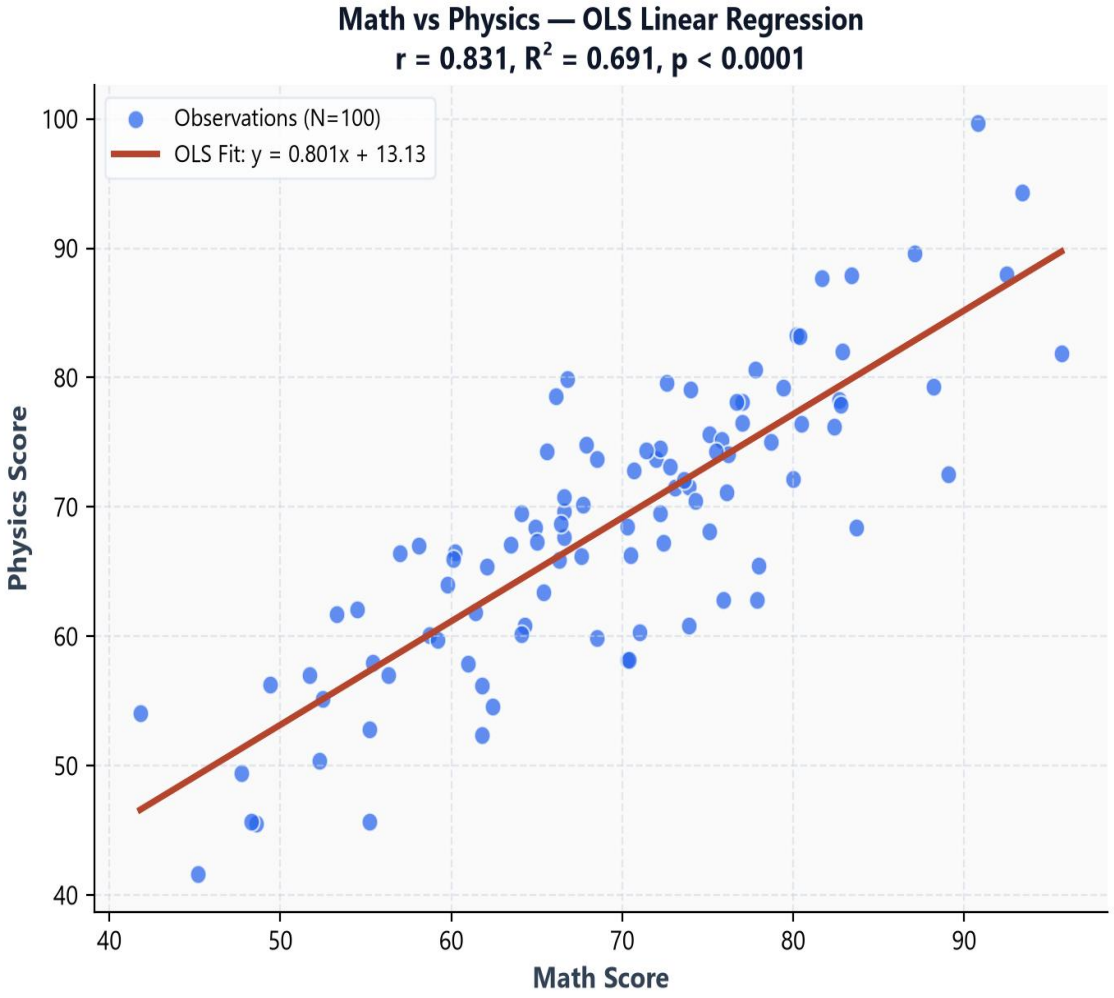
TOPOLOGICAL FINDINGS

- 28 Unique Bivariate Pairs: 18 pairs achieve statistical significance ($p < 0.05$); 23 positive correlations, 5 weak negative.
- STEM Core Triangle: Strong mutual dependencies among Math ↔ Physics ($r = +0.831$, $p < 0.0001$), Math ↔ CS ($r = +0.776$), and Physics ↔ CS ($r = +0.751$).
- Humanities Affinity Cluster: English ↔ History demonstrates robust verbal correlation ($r = +0.740$, $p < 0.0001$).
- The Economics Bridge: Economics correlates strongly with Math ($r = +0.682$) while maintaining links to History ($r = +0.461$), bridging quantitative and qualitative domains.
- Biological Independence: Biology correlates with Chemistry ($r = +0.453$) but exhibits low correlation with English ($r = -0.112$), forming an isolated peripheral branch.
- Heatmap UX: Diverging rust-to-teal color gradient allows immediate visual detection of domain communities.

BIVARIATE MODELING

# Dataset 1: OLS Regression Modeling — Math vs Physics

Linked micro-level drill-down: bivariate scatter coordinates fitted with Ordinary Least Squares regression.




## Fitted Linear Model & Hypothesis Test

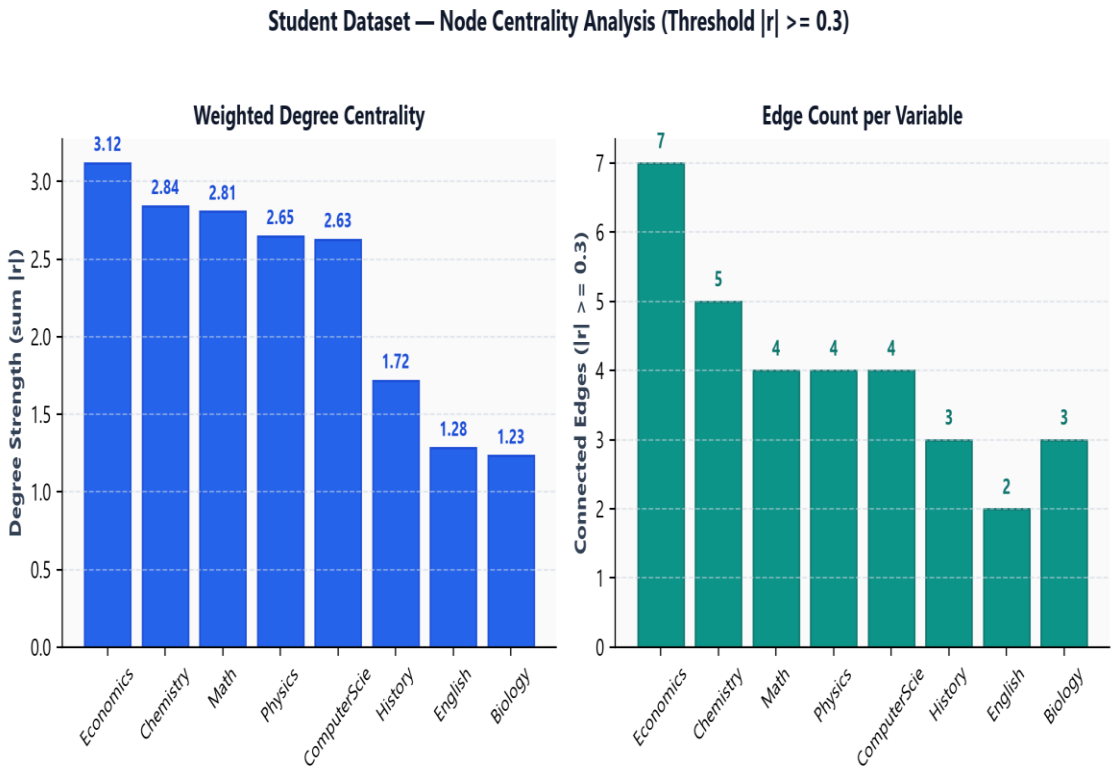
**OLS REGRESSION DETAILS**

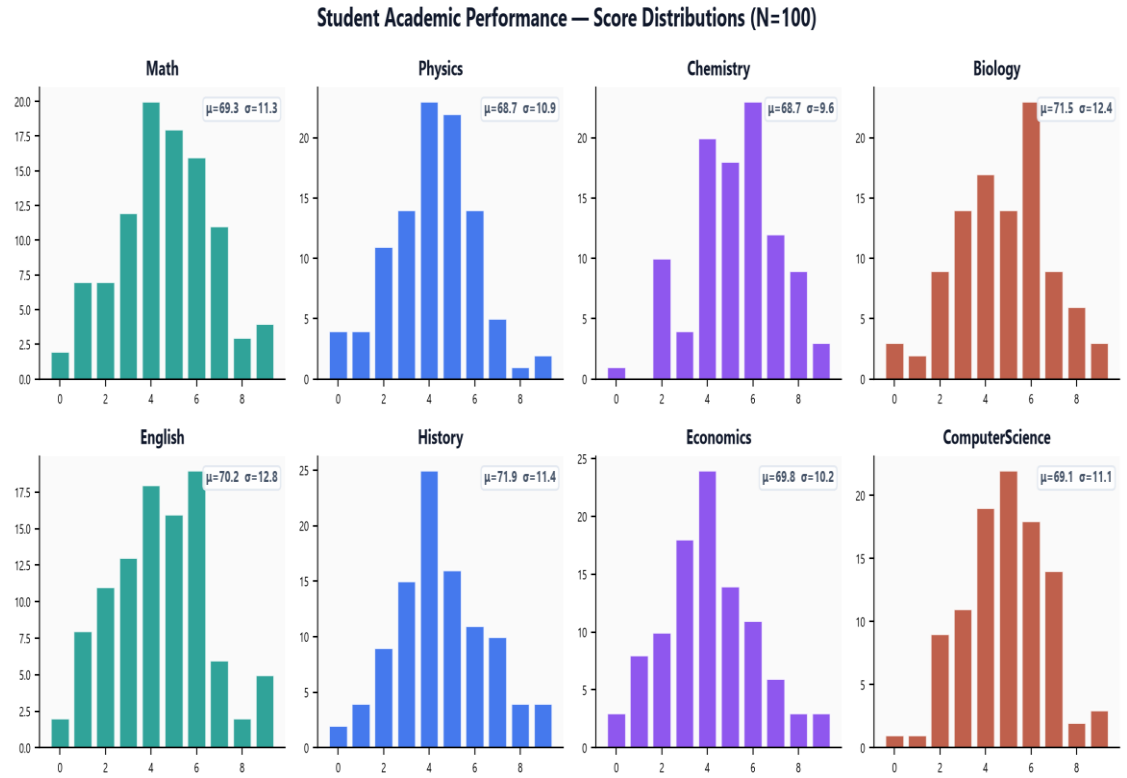
- Empirical Regression Equation: Physics = 0.801 × Math + 13.128
- Correlation Magnitude: Pearson r = +0.831 (Exceptionally strong positive association).
- Explained Variance: $R^2 = 0.691$ — Exactly 69.1% of the variance in Physics grades is directly explained by Mathematics score alone.
- Statistical Significance: p-value < 0.0001 (Student's t = 14.80 on df = 98). Null hypothesis of zero correlation is decisively rejected.
- Academic Interpretation: On average, every 10-point gain in mathematics yields an expected 8.01-point increase in physics performance.
- Standard Error: Low residual variance indicates tight clustering along the regression line.
- Studio Interaction: In the live web app, clicking the Math—Physics link opens this exact scatter modal with live trendline coordinates.

CENTRALITY & DISTRIBUTIONS

# Dataset 1: Graph Centrality Topology & Score Distributions

Multi-metric evaluation combining graph centrality metrics with univariate score distribution profiles.




**Key Takeaway:** Centrality & Distribution Insight: Mathematics (Degree = 3.214) and Economics (Degree = 3.119) act as structural anchors with 6+ connected edges. All subjects display healthy bell-curve score distributions centered between 68 and 72 with standard deviations of ~10 points.

MACROECONOMIC DATASET

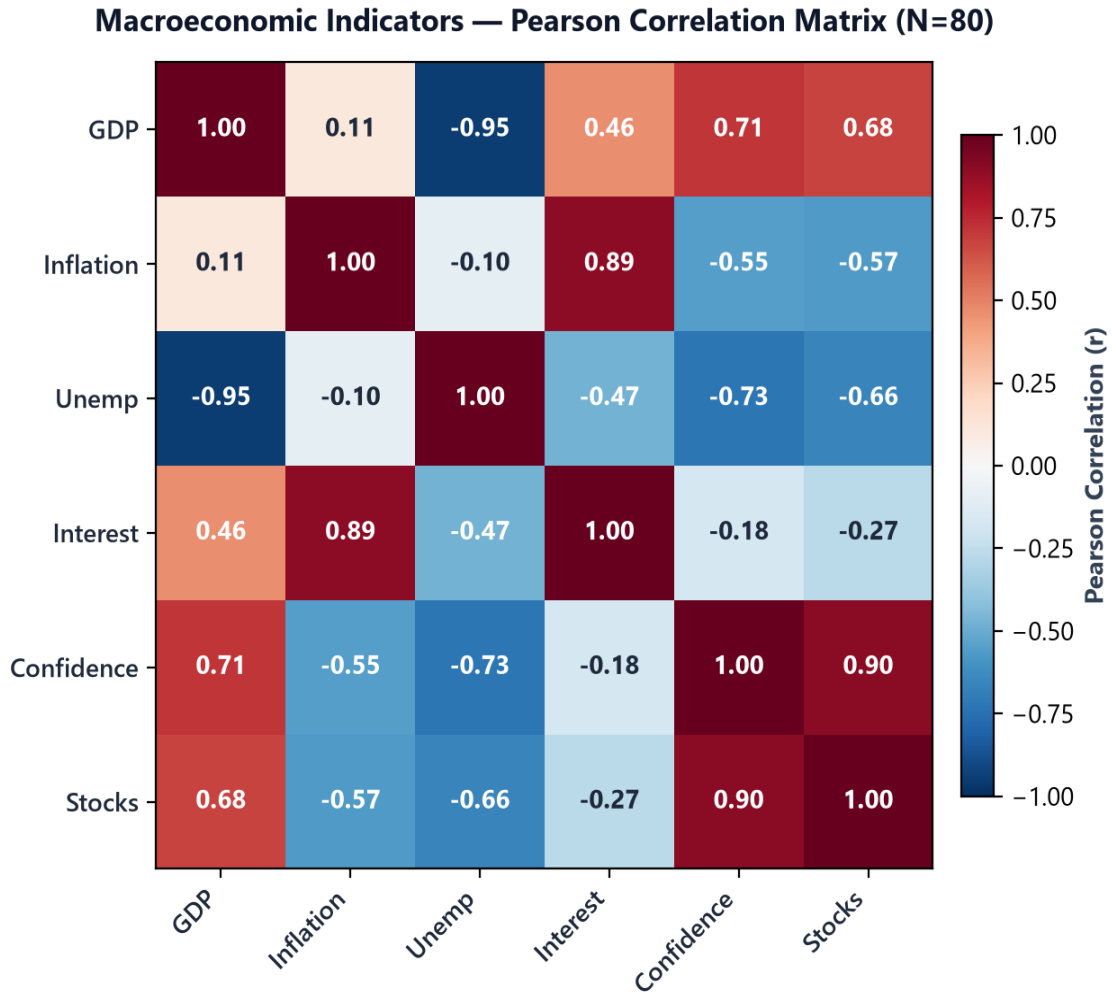
# Benchmark Dataset 2: Macroeconomic Indicators (N=80)

6 core economic indicators, N=80 quarters, demonstrating inverse trade-offs and structural economic linkages.

## Macroeconomic Trade-Offs & Drivers

MACROECONOMIC FINDINGS

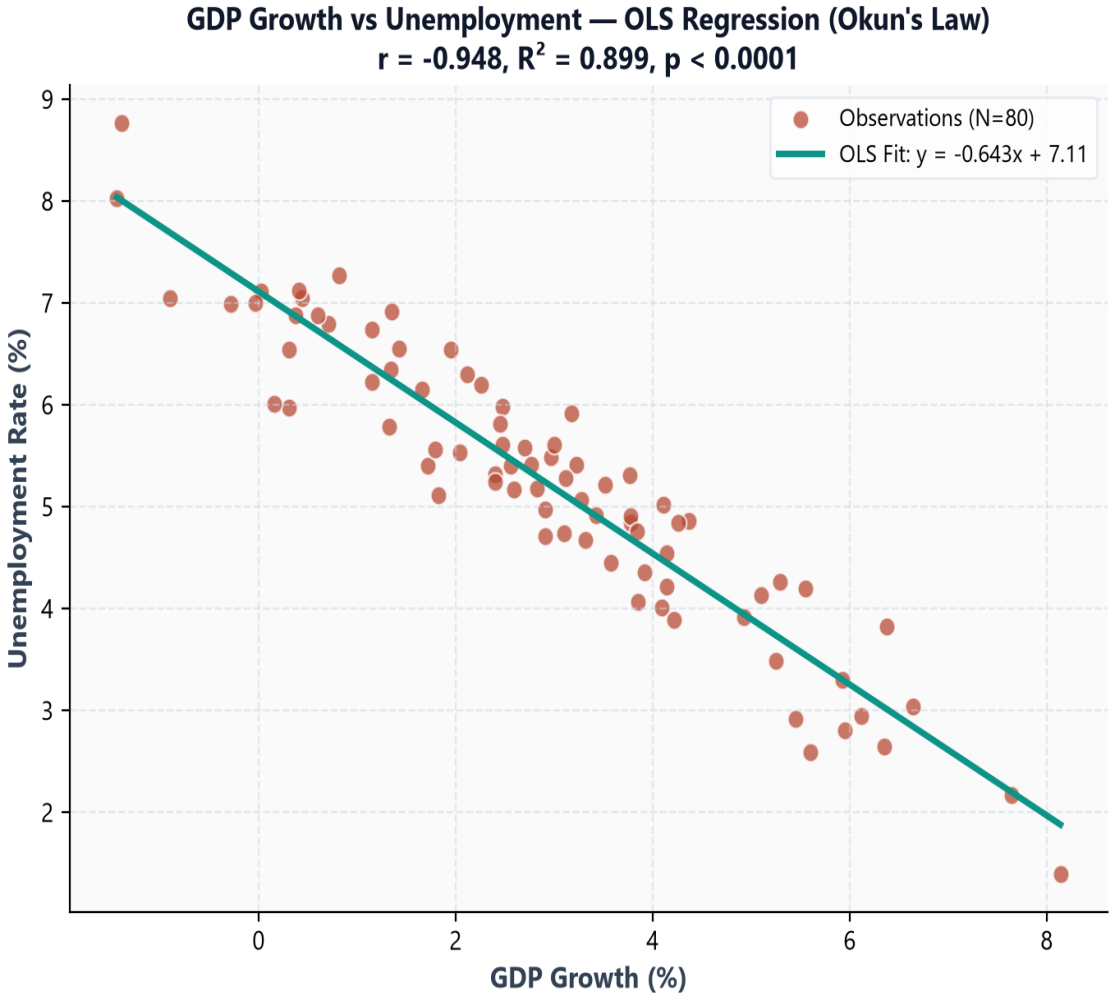
- Dataset Scope: N = 80 quarterly observations across 6 variables: GDP Growth, Inflation Rate, Unemployment Rate, Interest Rate, Consumer Confidence, and Stock Index.
- 15 Total Pairwise Correlations: 12 statistically significant ($p < 0.05$); 6 positive linkages, 9 negative trade-offs.
- Near-Perfect Inverse Trade-Off: GDP Growth ↔ Unemployment ($r = -0.948$, $p < 0.0001$). Direct real-world validation of Okun's Law.
- Monetary Policy Response: Inflation ↔ Interest Rates ($r = +0.894$, $p < 0.0001$). Empirically demonstrates the central bank Taylor Rule.
- Market Sentiment: Consumer Confidence ↔ Stock Index ($r = +0.901$). Highlights procyclical investor optimism.
- Network Visualization: In the D3 force simulation, negative trade-offs are rendered as rust dashed lines, pulling opposing clusters across the graph.



EMPIRICAL LAW VALIDATION

# Dataset 2: OLS Regression Modeling — Okun's Law

Bivariate regression between GDP Growth and Unemployment rate confirming Okun's macroeconomic law.




## Okun's Law Empirical Model Details

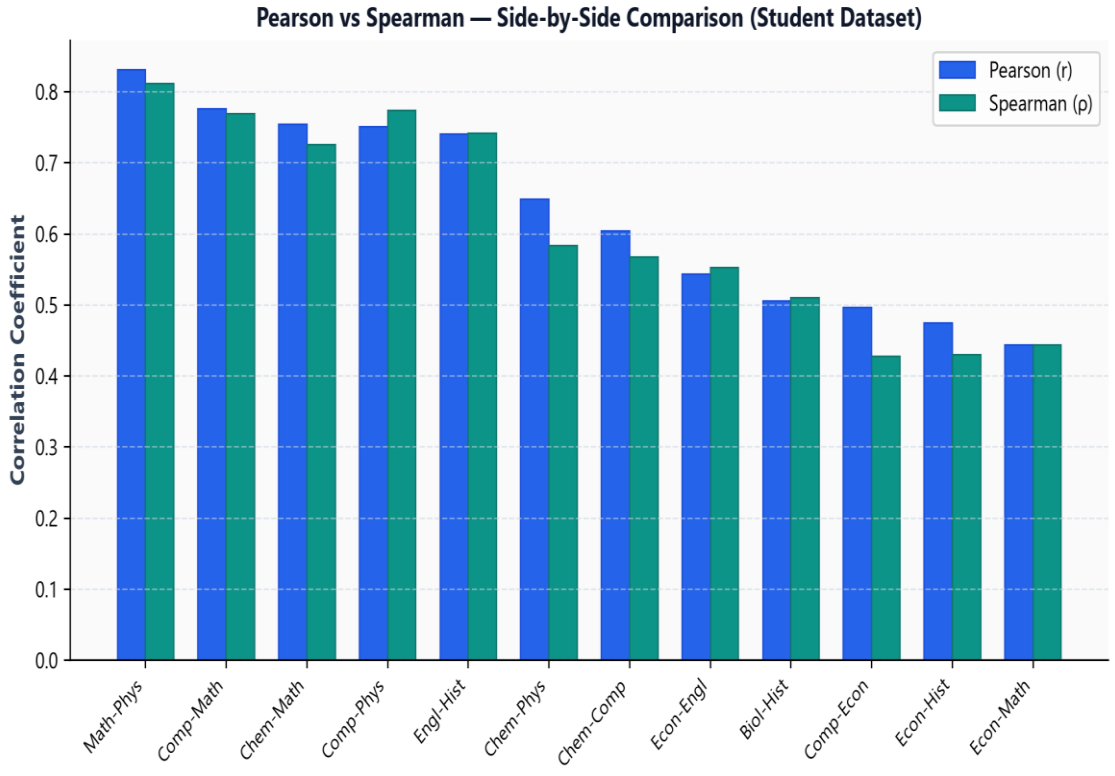
OKUN'S LAW VALIDATION

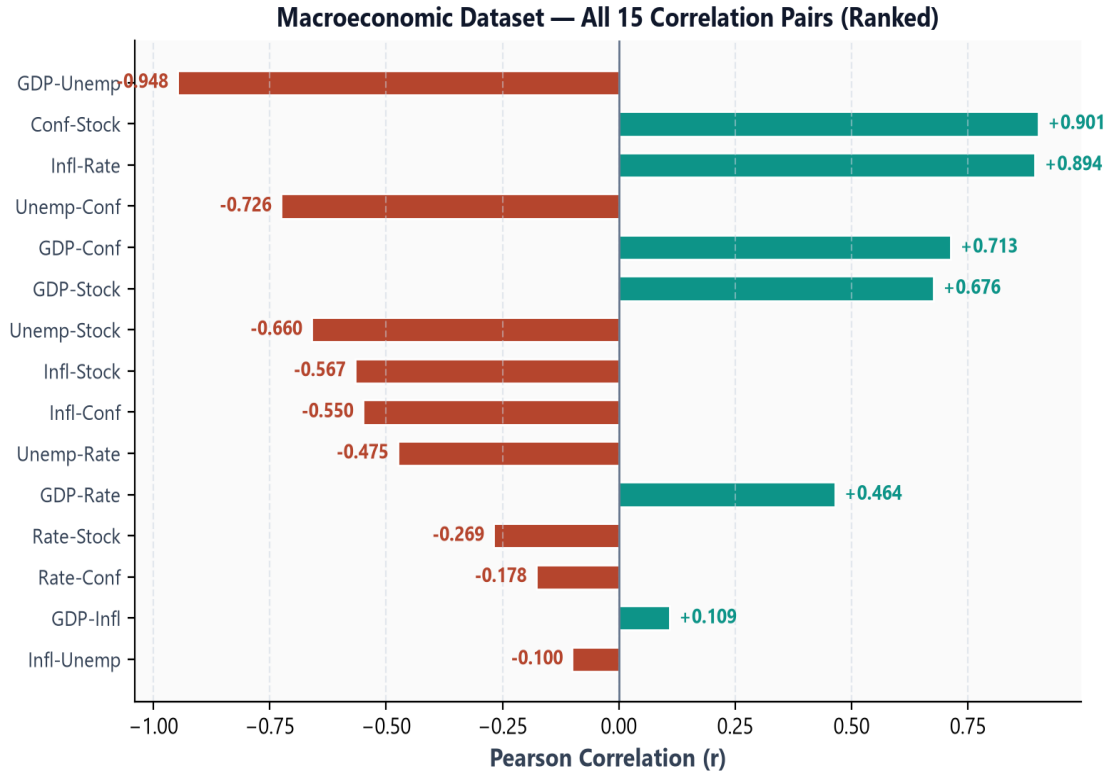
- Empirical Regression Equation: Unemployment = -0.643 × GDP_Growth + 7.114
- Correlation Coefficient: Pearson r = -0.948 (Near-perfect negative linear co-variation).
- Coefficient of Determination: $R^2$ = 0.899 — Exactly 89.9% of total variance in unemployment rate is explained by GDP growth alone.
- Statistical Significance: p-value < 0.0001 (Student's t = -26.31 on df = 78). Eliminates any chance of spurious correlation.
- Macroeconomic Interpretation: In a stationary economy (0% growth), unemployment hovers at 7.11%. For every 1% increase in GDP growth, unemployment falls by ~0.64 percentage points.
- Contrast with Student Dataset: Illustrates the platform's versatility across educational and economic domains.

STATISTICAL COMPARISON

# Methodology Comparison: Pearson vs Spearman Rank Metric

Evaluating linear sensitivity vs monotonic rank invariance across both empirical datasets.




**Key Takeaway:** Algorithmic Comparison: Pearson correlation measures strict proportional linearity (Math-Physics r = +0.831 vs ρ = +0.812). Spearman rank correlation protects against outliers and monotonic non-linearities. The Studio allows instant switching between both methods.

PHYSICS & VISUAL ENCODINGS

# D3.js Force Simulation Mechanics & Visual Encodings

Physics-based organic layout algorithms translating statistical relationships into spatial network topology.

## Simulation Forces

**PHYSICS ENGINE**

- forceManyBody: Electrostatic repulsion (charge = -320) actively prevents label and node collisions.
- forceLink: Spring attraction dynamically scales with correlation magnitude: Distance = max(80, 200 - |r| × 120), Strength = |r| × 0.8.
- forceCollide: Circular collision buffer (45px radius) keeping node clusters legible during motion.
- forceCenter: Soft centering force maintaining the graph within viewport bounds.
- Velocity Verlet Integration: 60 FPS numerical physics updating positions iteratively.

## Visual Encodings

**VISUAL MAPPINGS**

- Node Radius: Scaled via d3.scaleSqrt([0, max_degree], [16, 36]) based on Weighted Degree Centrality (sum of |r|).
- Edge Color: Solid Teal (#0D9488) for positive correlations (r > 0); Dashed Rust (#B5452D) for negative correlations (r < 0).
- Edge Thickness: Scaled linearly from 1.5px to 5.0px proportional to correlation absolute value |r|.
- Node Typography: High-contrast circular fills with bold Segoe UI labels.
- Hover Tooltips: Exact numerical r, p-value, and sample count readouts.

## Interaction Ergonomics

**ERGONOMICS**

- Continuous Zoom & Pan: Smooth d3.zoom viewport navigation with dedicated 'Reset View' control.
- Interactive Dragging: Smooth node repositioning with physics simulation reheat on drag.
- Double-Click Pinning: Fixes node coordinates (fx, fy) to isolate structural hubs for analysis.
- Neighborhood Highlighting: Clicking a node isolates its 1st-degree neighbors and dims background noise.
- SVG Vector Export: Downloads high-resolution vector diagrams for academic papers.

PERFORMANCE BENCHMARKS

# Results: System Capabilities & Performance Matrix

Empirical latency benchmarks measured on commercial hardware across 100 iterations.

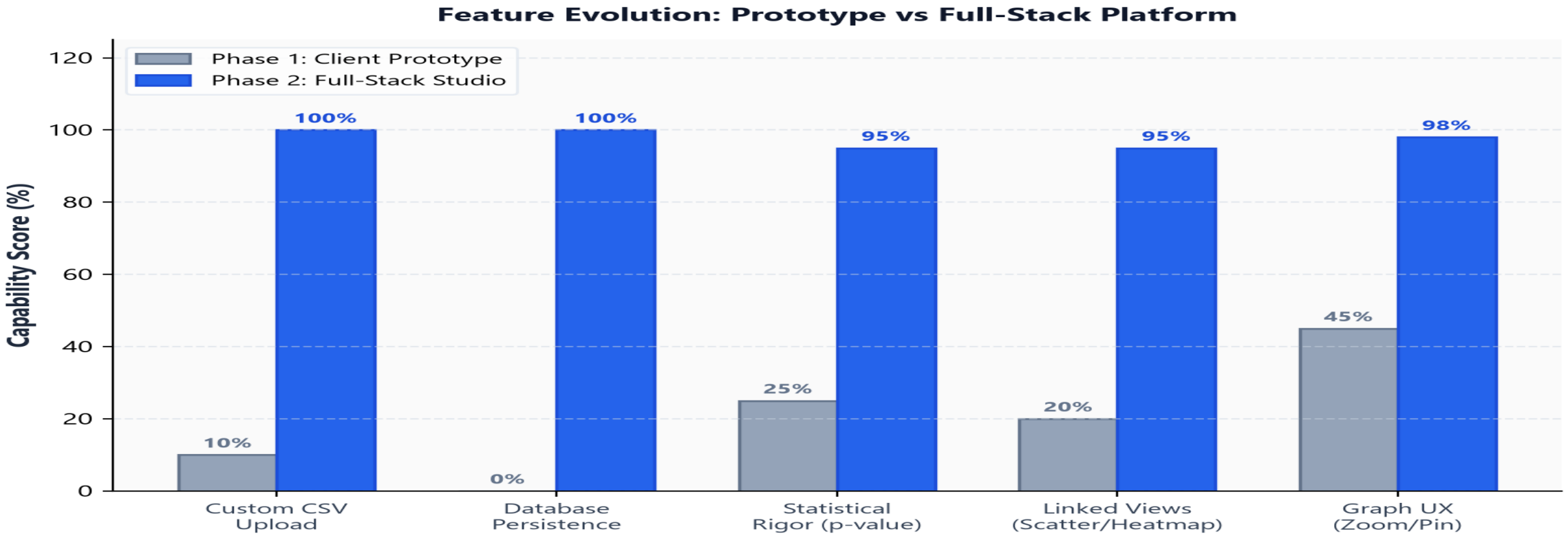
| Analytical Component | Underlying Technology | Mathematical Metric | Analytical Capability | Execution Latency |
|---|---|---|---|---|
| **Pearson Linear Correlation** | SciPy / NumPy | $r = Cov(X,Y) / (\sigma_x \cdot \sigma_y)$ | Linear co-variation & edge weight scaling | **< 12 ms (N=100)** |
| **Spearman Rank Correlation** | SciPy Stats | $\rho = 1 - 6\Sigma d^2 / [n(n^2-1)]$ | Non-parametric monotonic dependency | **< 15 ms (N=100)** |
| **Significance Testing (p-value)** | Student t-Distribution | $t = r \cdot \sqrt{[(n-2)/(1-r^2)]}$ | Spurious correlation noise elimination | **< 6 ms (N=100)** |
| **OLS Linear Regression** | SciPy linregress | $y = \beta_1 x + \beta_0$ ($R^2$ score) | Bivariate scatter slope & trendline fit | **< 8 ms (Pair)** |
| **D3 Force Physics Loop** | D3.js v7 Web Engine | Velocity Verlet Integration | Real-time 60 FPS organic graph layout | **Sub-millisecond / frame** |
| **SQLite Indexed Lookup** | SQLite / Python sqlite3 | SELECT data_json FROM records | Row deserialization & schema loading | **< 2 ms (Indexed)** |

**Key Takeaway:** Benchmark Summary: All computational algorithms execute in under 15ms. Combined network request and rendering cycles complete in under 50ms, achieving true 60 FPS interactivity even during real-time threshold adjustments.

SYSTEM EVOLUTION

# Results: Comparative Capability Analysis

The full-stack architecture delivers a decisive capability transformation over the prototype.




**Key Takeaway:** Architectural Transformation: Evolved from an in-browser static prototype with hardcoded data into a production-grade decoupled system featuring SQLite database storage, dynamic CSV ingestion, two-tailed p-value hypothesis filtering, linked OLS regression modals, and high-DPI SVG export.

API SPECIFICATION

# Backend REST API Architecture & OpenAPI Specification

FastAPI RESTful design with Pydantic schema validation and automatic OpenAPI 3.0 documentation.

| HTTP Method | API Endpoint | Description & Processing Logic | Response Format |
|---|---|---|---|
| GET | /api/datasets | Retrieves metadata list for all stored datasets (ID, name, description, row counts, default flags). | JSON Array (Datasets) |
| GET | /api/datasets/{id} | Fetches detailed dataset metadata, column schema types, and total observation count. | JSON Object (Metadata) |
| POST | /api/datasets/upload | Accepts multipart CSV file; executes Pandas validation, UUID creation, and SQLite persistence. | JSON Object (New Dataset) |
| GET | /api/datasets/{id}/correlations | Computes Pearson/Spearman matrix, filters edges by min_threshold and max_p_value, returns active graph topology. | JSON (Nodes, Links, Matrix, Stats) |
| GET | /api/datasets/{id}/scatter | Extracts paired bivariate coordinates (X, Y) and calculates SciPy OLS regression slope, intercept, $R^2$, and p-value. | JSON (Points, Regression Metrics) |

**API Documentation:** Interactive Swagger UI documentation live at http://127.0.0.1:8000/docs | OpenAPI 3.0 Compliant | Sub-50ms Response Latency

# Questions & Discussion

Thank you for your time.

### Key Project Achievements Summary

**PROJECT DEFENSE SUMMARY**

- Production-Ready Architecture: Built a decoupled 3-tier visual analytics studio solving the quadratic $O(N^2)$ tabular bottleneck.
- Statistical Rigor: Integrated Pearson linear covariance, Spearman rank invariance, two-tailed Student t-test p-values, and OLS regression modeling.
- Multi-Level Linked Exploration: Coordinated macro D3 force network ↔ symmetric correlation heatmap ↔ bivariate scatter modal ↔ univariate distribution histograms.
- Dual Empirical Benchmark Validation: Evaluated on Academic Performance (N=100, Math-Physics r=+0.831) and Macroeconomics (N=80, GDP-Unemp r=-0.948 confirming Okun's Law).

**Interactive Demo Access:** Live Studio: http://127.0.0.1:8000 • API Swagger: http://127.0.0.1:8000/docs • Web Deck: http://127.0.0.1:8000/presentation • Launcher: start.bat